

Answer **all** questions.

1. (a) A student made some copper(II) sulfate crystals by reacting copper(II) carbonate powder with sulfuric acid using the following method.

Stage 1      Measure 50 cm<sup>3</sup> of sulfuric acid into a beaker.

Stage 2      Add copper(II) carbonate powder, one spatula at a time, until all the acid has reacted.

Stage 3      Filter the mixture.

Stage 4      Obtain crystals from the solution.

- (i) State how you would carry out Stage 4 to get the largest possible crystals. [1]

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- (ii) Crystals of copper(II) sulfate could also be made using copper(II) oxide powder instead of copper(II) carbonate powder. State and explain how the observations in Stage 2 would be different. [2]

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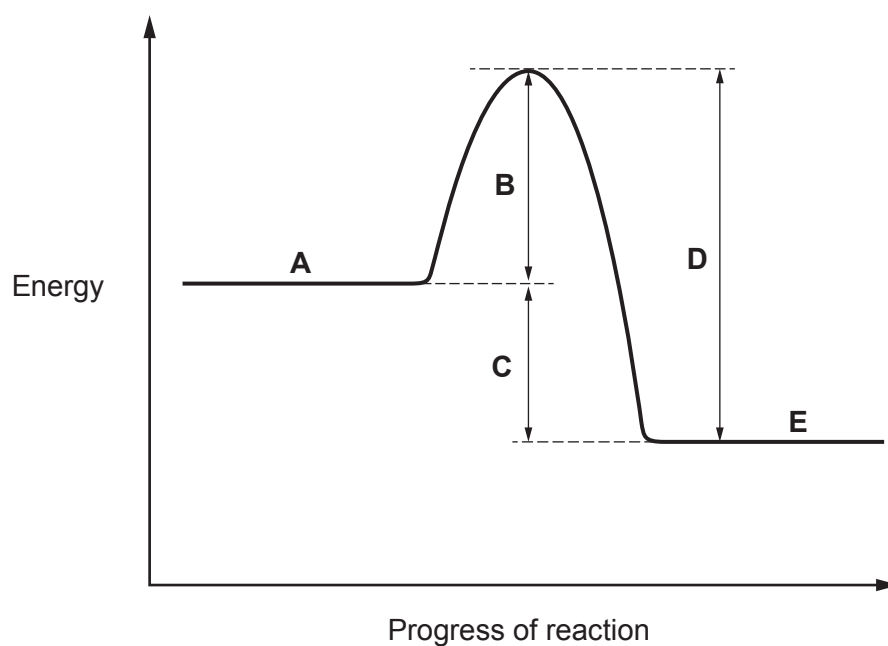
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- (iii) Complete the symbol equation for the reaction between copper(II) oxide and sulfuric acid. Copper(II) sulfate is one of the products. [2]



(b) The diagram shows an energy profile for a reaction.



(i) Give the **letter** that represents each of the following parts of the energy profile. [2]

| Part of the energy profile        | Letter |
|-----------------------------------|--------|
| energy change for the reaction    |        |
| energy of the reactants           |        |
| activation energy of the reaction |        |

(ii) Give the meaning of the term activation energy. [1]

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(iii) State how the energy profile shows that this is an exothermic reaction. [1]

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